

SHIPNOBLE

Transportation Division

ShipNoble Index (SNI) — Phase Architecture

Straight Truck Operations • Palletized Freight • Profitability Framework

Asset Rate	Operator Rate	Add'l Pallet	Fixed Truck	Target Margin
\$0.55 / km	\$0.7462 / min	\$75.00 flat	\$156.59 / day	40% (SNI 1.0)

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Executive Summary

This document defines the ShipNoble Index (SNI) for the Transportation Division — a complete profitability framework that measures margin performance at every level of operations, from individual pallets to full manifests, from specific customer accounts to geographic destinations, and from dock efficiency to operator execution.

The framework is built on a single formula:

$$\text{SNI} = ((\text{Revenue} - \text{Cost}) / \text{Revenue}) + 0.6$$

An SNI of 1.0 equals the 40% target margin. An SNI of 0.6 is break-even. Every 0.1 increment represents exactly 10 percentage points of margin movement. This standardized scale allows direct comparison across every dimension of the business.

Revenue Model

Every customer is charged as though they are receiving a dedicated round-trip delivery from the terminal. The pricing schedule is derived from a cost-plus model that targets 40% margin on a standalone basis. When multiple customers are served on a single optimized route, each still pays their full standalone price. The spread between collected revenue and shared route cost is the structural profit advantage of this division.

The pricing schedule is bidirectional — pickups and deliveries are priced identically, because the truck makes the same round-trip regardless of freight direction.

Cost Architecture

Costs are allocated using three operational buckets (stem overhead, inter-stop leg cost, and dwell cost) plus a daily fixed truck cost. Stem overhead is allocated proportional to the time each stop consumes on the route. The daily fixed truck cost of \$156.59 is allocated proportional to pallet count, ensuring that stops generating more revenue absorb proportionally more overhead. Dwell cost uses only the operator rate, keeping facility friction cleanly isolated.

Measurement Hierarchy

The SNI is calculated at six levels: Pallet SNI (unit economics), Consignment SNI (booking profitability), Shipper SNI (account health), Destination SNI (geographic performance across 45 distance tiers), Dock SNI (facility friction), and Manifest SNI (route-level profitability). An Operator SNI measures execution variance between predicted and actual performance.

Key Insight

Structural Profitability

The only scenario in which the Transportation Division loses money on a delivery is a single-pallet, single-stop run where actual costs exceed the priced cost by more than 40%. The cost-plus pricing model prevents this by design. Every multi-stop manifest generates margin above the 40% baseline because customers collectively pay multiple standalone prices while the truck runs one shared route.

1. Movement Classification

Every straight truck movement is classified into one of four types before any financial calculation begins. The classification determines which revenue and cost formulas apply.

Movement Type	Revenue	SNI Scope	Cost Treatment
Inbound / Feeder Sweep	Yes	Standalone	Full round-trip cost self-contained; piece-rate revenue at \$2.00 per CE collected
Internal Transfer	None	Cost Center	Full cost allocated to the department that requested the transfer
Outbound Manifest	Yes	Full Hierarchy	Three-bucket allocation plus fixed truck cost; full SNI hierarchy applies
Linehaul / Deadhead	None	Cross-Division	Cost adjusted by Impact Multiplier, allocated to destination department

1.1 Feeder Sweeps

Single-leg out-and-back movements collecting product from supplier facilities. Revenue equals the number of Case Equivalents collected multiplied by \$2.00. Cost is the full round-trip at standard rates. Each sweep carries its own standalone SNI.

1.2 Internal Transfers

Movements between ShipNoble-controlled facilities. These generate no external revenue. The requesting department absorbs the full cost, maintaining clear accountability and preventing internal logistics from invisibly eroding manifest profitability.

1.3 Outbound Manifests

Multi-stop delivery and pickup runs serving external customers. This is the core profit engine and the only movement type that flows through the complete SNI hierarchy. All subsequent sections of this document apply to Outbound Manifests.

1.4 Linehauls and Deadheads

Hub-to-hub round-trips where the Transportation Division acts as an internal service provider. An Impact Multiplier adjusts the allocated cost based on whether the return leg carries revenue freight.

$$\text{Impact Multiplier (IM)} = (\text{Gross_Cost} - \text{Backhaul_Revenue}) / \text{Gross_Cost}$$

No return freight: IM = 1.00 (full cost to destination department)
 Partial backhaul: IM < 1.00 (offset by return revenue)
 Full backhaul: IM = 0.00 (cost-neutral round-trip)

$$\text{Net Allocated Cost} = \text{Gross_Cost} \times \text{IM}$$

2. Pricing Model

The distance-tier pricing schedule is derived from a cost-plus methodology. Each destination's first-pallet price is engineered to produce exactly 40% gross margin on a standalone round-trip delivery.

2.1 Price Derivation

```
First Pallet Price = Cost_to_Serve / 0.60, rounded to nearest $5
```

```
Cost_to_Serve = Round-Trip Transit Cost + Fixed Truck Share
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```
Round-Trip Transit Cost = (RT_km × $0.55) + (RT_min × $0.7462)
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```
Fixed Truck Share = $156.59 / assumed stops per manifest
```

```
Dwell time is excluded from the pricing basis.
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```
Each additional pallet at the same stop: $75.00 flat.
```

```
Minimum shipment: 1 pallet.
```

Dividing by 0.60 is the mathematical complement of a 40% target margin. A cost of \$60 divided by 0.60 yields a price of \$100, producing a \$40 margin at exactly 40%. This means an SNI of 1.0 is built into every price by construction.

2.2 Bidirectional Application

The pricing schedule applies equally to pickups and deliveries. The truck makes the same round-trip regardless of freight direction, so the cost basis is identical. A pickup from Cambridge is priced the same as a delivery to Cambridge.

2.3 Revenue on Multi-Stop Manifests

Each customer on a manifest is invoiced their full standalone round-trip price. A five-stop manifest collects five full prices. The customer is unaware that other stops exist on the route. Revenue therefore scales linearly with the number of stops, while the actual cost of operating one optimized route scales sub-linearly.

Density Economics

A 3-stop manifest collecting \$970 in standalone prices against \$383 in shared route cost produces a 60.5% margin (SNI 1.205). The 20.5% margin above the 40% target is pure density profit from route consolidation. Every additional stop on a manifest widens this spread.

2.4 Revenue Formula

```
Per stop i (delivery or pickup):
```

```
R_i = TierPrice(destination_i) + max(0, pallets_i - 1) × $75.00
```

```
Manifest Revenue:
```

```
R_manifest = ∑ R_i for all stops on the manifest
```

3. Cost Allocation

The total cost of each stop on an Outbound Manifest is the sum of four components, each with a specific allocation method designed to match cost to the resource being consumed.

$$C_{\text{stop}} = C_{\text{stem_share}} + F_{\text{truck_share}} + C_{\text{leg}} + C_{\text{dwell}}$$

3.1 Stem Overhead — Time-Proportional

The stem is the non-revenue-generating movement from the terminal to the first stop and from the last stop back to the terminal. This overhead is allocated proportional to each stop's share of total manifest stop time. Stops that consume more route time bear more overhead.

$$\begin{aligned} \text{Stem First (SCF)} &= (d_{\text{first}} \times \$0.55) + (t_{\text{first}} \times \$0.7462) \\ \text{Stem Last (SCL)} &= (d_{\text{last}} \times \$0.55) + (t_{\text{last}} \times \$0.7462) \\ C_{\text{stem}} &= \text{SCF} + \text{SCL} \\ \text{Allocation: } C_{\text{stem_share},i} &= C_{\text{stem}} \times (t_{\text{stop},i} / T_{\text{all_stops}}) \\ t_{\text{stop},i} &= \text{transit time to stop } i + \text{dwell time at stop } i \\ T_{\text{all_stops}} &= \text{sum of } t_{\text{stop}} \text{ for all stops (excludes stem)} \end{aligned}$$

3.2 Fixed Truck Cost — Pallet-Proportional

The daily cost of operating the truck (\$3,445 per month divided by 22 operating days = \$156.59 per day) covers insurance, depreciation, licensing, parking, and maintenance. This cost is allocated proportional to pallet count, ensuring that stops generating more revenue (more pallets) absorb proportionally more fixed overhead.

$$\begin{aligned} F_{\text{truck}} &= \$3,445 / 22 = \$156.59 \text{ per operating day} \\ \text{Allocation: } F_{\text{share},i} &= F_{\text{truck}} \times (\text{pallets}_i / N_{\text{pallets}}) \\ N_{\text{pallets}} &= \text{total pallets across entire manifest} \\ \text{Multi-manifest days:} \\ F_{\text{manifest}} &= F_{\text{truck}} \times (T_{\text{this_manifest}} / T_{\text{operating_day}}) \\ &\text{then pallet-proportional within each manifest} \end{aligned}$$

3.3 Inter-Stop Leg Cost — Direct to Receiving Stop

The cost of driving from one stop to the next is allocated entirely to the destination stop. The first stop carries zero leg cost because the terminal-to-first-stop movement is captured in the stem overhead.

$$\begin{aligned} C_{\text{leg},i} &= (d_{\text{leg},i} \times \$0.55) + (t_{\text{leg},i} \times \$0.7462) \\ C_{\text{leg},1} &= \$0 \quad (\text{terminal to first stop is in stem}) \end{aligned}$$

3.4 Dwell Cost — Operator Rate Only

Time spent at a customer's dock is costed at the operator rate only. The asset rate does not apply because the truck is stationary. This deliberate separation keeps dwell cost in the time domain, enabling clean isolation of dock friction in the Dock SNI metric. The asset's opportunity cost while parked is captured through the fixed truck allocation.

```
C_dwell,i = t_dwell,i × $0.7462
```

Dwell is not included in the pricing basis.
Every minute of dwell is pure cost with no offsetting revenue.
This makes dock efficiency a direct lever on profitability.

3.5 Multi-Consignment Stops

When two or more customers share a single stop, the stop cost is split proportional to each customer's pallet count. This applies to all four cost components.

```
C_consignment = C_stop × (consignment_pallets / total_pallets_at_stop)
```

Example: 3 pallets at one stop (2 for Customer A, 1 for Customer B)
Customer A bears 66.7% of stop cost
Customer B bears 33.3% of stop cost

3.6 Worked Example

A three-stop manifest departing ACC Terminal, carrying 8 total pallets.

Revenue

Stop	Destination	Pallets	1st Pallet	Additional	Stop Revenue
S1	Toronto	3	\$130.00	$2 \times \$75 = \150	\$280.00
S2	Vaughan	1	\$165.00	—	\$165.00
S3	Hamilton	4	\$300.00	$3 \times \$75 = \225	\$525.00

Manifest Revenue: \$280 + \$165 + \$525 = \$970.00

Route Cost Breakdown

Segment	km	min	Asset	Operator	Total	Bucket
Stem: Terminal → S1	18	22	\$9.90	\$16.42	\$26.32	Stem
Dwell at S1 (3 pallets)	—	25	—	\$18.66	\$18.66	Dwell
Leg: S1 → S2	28	24	\$15.40	\$17.91	\$33.31	Leg → S2
Dwell at S2 (1 pallet)	—	15	—	\$11.19	\$11.19	Dwell
Leg: S2 → S3	45	38	\$24.75	\$28.36	\$53.11	Leg → S3
Dwell at S3 (4 pallets)	—	30	—	\$22.39	\$22.39	Dwell
Stem: S3 → Terminal	52	44	\$28.60	\$32.83	\$61.43	Stem

Allocation to Stops

```

STEM (time-proportional):  C_stem = $87.75
  S1: 25 min / 132 = 18.94% → $16.62
  S2: 39 min / 132 = 29.55% → $25.93
  S3: 68 min / 132 = 51.52% → $45.21

FIXED TRUCK (pallet-proportional):  F_truck = $156.59
  S1: 3 / 8 = 37.50% → $58.72
  S2: 1 / 8 = 12.50% → $19.57
  S3: 4 / 8 = 50.00% → $78.30

```

Stop	Stem Share	F_truck Share	Leg Cost	Dwell Cost	Total Cost	Revenue
S1	\$16.62	\$58.72	\$0.00	\$18.66	\$94.00	\$280.00
S2	\$25.93	\$19.57	\$33.31	\$11.19	\$90.00	\$165.00
S3	\$45.21	\$78.30	\$53.11	\$22.39	\$199.01	\$525.00

```

C_manifest = $94.00 + $90.00 + $199.01 = $383.01
R_manifest = $970.00
Profit      = $586.99
Margin      = 60.5%
SNI         = (0.605) + 0.6 = 1.205

```

4. Evaluation States

Every manifest is evaluated in two states. The formulas are identical — only the data source changes. Revenue remains constant across both states.

	Scheduled (Pre-Dispatch)	Realized (Post-Run)
Time & Distance	Google Maps API predictions	DMS system actual data
Dwell Prediction	10 min base + 5 min per pallet	DMS: departure – arrival timestamp
Stop Sequence	Planned order	Actual order (system auto-repopulates)
Purpose	Route viability, dispatch decisions	Performance review, accountability

4.1 Dwell Prediction

Since Google Maps provides only transit time, dwell is predicted using a pallet-based formula: 10 minutes of base time (arrival, check-in, paperwork, departure) plus 5 minutes per pallet for physical handling. This prediction serves as both the Scheduled cost input and the baseline for Dock SNI measurement.

Dwell time is not included in the pricing basis. It is a cost-only item, meaning every minute at the dock erodes margin without corresponding revenue. This makes dock efficiency one of the most direct levers on realized profitability.

4.2 Sequence Deviation Handling

If an operator visits stops in a different order than planned, the system automatically re-sequences the manifest and recalculates a new Scheduled SNI using Google Maps data for the actual order. This ensures the variance comparison between Scheduled and Realized is always against the same stop sequence.

5. SNI Hierarchy

The SNI is calculated at six levels, organized into three arms. Each level uses the same master formula with inputs scoped to the relevant business dimension.

$$\text{SNI} = ((R - C) / R) + 0.6$$

1.0 = 40% margin (target)

0.6 = break-even

Each 0.1 = 10% margin

$$\text{Margin\%} = (\text{SNI} - 0.6) \times 100$$

5.1 Commercial Arm

Pallet SNI

The atomic unit of profitability. Cost is allocated to each pallet proportional to its revenue contribution. The first pallet (at the distance-tier price) bears proportionally more cost than additional pallets (at \$75), producing comparable SNIs across pallet types at the same stop.

Consignment SNI

A consignment is one customer's freight at one stop. Revenue is the first-pallet price plus \$75 per additional pallet. Cost is the full stop cost (or pallet-weighted share if multiple consignments share the stop). More pallets per consignment means higher revenue against the same stop cost, rewarding freight density.

Shipper SNI

The aggregated profitability of a paying customer across all their consignments in a rolling 30-day window. Calculated as a single pooled computation: sum all revenue, sum all costs, apply the formula once. This provides an at-a-glance measure of account health.

5.2 Spatial Arm

Destination SNI (45 Distance Tiers)

One SNI per pricing schedule destination, rolling 30-day window. Since every price is engineered for 40% margin, a destination trending below 1.0 signals either operational inefficiency (dwell, routing) or a cost basis that has drifted from reality. A destination consistently above 1.0 indicates efficiency or multi-stop density benefits.

5.3 Dock SNI (Facility Friction)

The Dock SNI isolates how a specific facility's dwell behaviour compares to what it should cost. It requires only three data points per visit: the DMS arrival timestamp, the DMS departure timestamp, and the pallet count.

$$\text{Expected dwell: } t_{\text{expected}} = 10 + (\text{pallets} \times 5) \text{ minutes}$$

```

Actual dwell:    t_actual    = DMS departure - DMS arrival

Per visit:      Dock SNI = ((t_expected - t_actual) / t_expected) + 0.6
Aggregate:      Dock SNI = ((Σ t_expected - Σ t_actual) / Σ t_expected) + 0.6

```

A Dock SNI of 0.60 means the facility matches prediction. Above 0.60 means faster than expected. Below 0.60 means the dock is adding friction — a score below 0.40 (20% slower than predicted) should trigger a review for surcharges or operational discussion with the facility.

5.4 Execution Arm

Manifest SNI

The route-level profitability metric. Calculated in both Scheduled and Realized states. The difference between these two values is the total execution variance.

Operator SNI

Measures execution quality by normalizing the variance between Scheduled and Realized Manifest SNI against the scheduled margin magnitude. An Operator SNI of 1.0 means the operator matched prediction exactly. Above 1.0 means they outperformed; below 1.0 means they underperformed. Normalization ensures comparability across routes of different revenue sizes.

```

Operator SNI = 1.0 + (Δ_manifest / |SNI_scheduled - 0.6|)

where Δ_manifest = SNI_realized - SNI_scheduled

```

6. Variance Analysis

The total difference between Scheduled and Realized Manifest SNI is decomposed into two components to assign accountability correctly.

6.1 Total Variance

$$\begin{aligned}\Delta_{\text{total}} &= \text{SNI}_{\text{realized}} - \text{SNI}_{\text{scheduled}} \\ &= (\text{C}_{\text{scheduled}} - \text{C}_{\text{realized}}) / \text{R}_{\text{manifest}}\end{aligned}$$

6.2 Dock Variance (Isolatable)

The cost impact of actual dwell times differing from predicted dwell times. This is entirely attributable to facility behaviour and is not within the operator's control.

$$\begin{aligned}\Delta_{\text{dock_cost}} &= \sum [(t_{\text{dwell_actual}} - t_{\text{dwell_predicted}}) \times \$0.7462] \\ \Delta_{\text{dock}} &= -\Delta_{\text{dock_cost}} / \text{R}_{\text{manifest}}\end{aligned}$$

6.3 Operator Variance (Residual)

Everything that remains after removing dock friction: driving speed relative to prediction, route deviations, construction delays, and stem time differences.

$$\Delta_{\text{operator}} = \Delta_{\text{total}} - \Delta_{\text{dock}}$$

Accountability

Dock variance is attributable to the customer facility. Operator variance is attributable to the driver. This separation prevents operators from being penalized for slow docks, and prevents facility problems from being hidden inside overall route performance.

Appendix A: Distance-Tier Pricing Schedule

Bidirectional: identical for pickups and deliveries. First pallet prices below; each additional pallet at same stop is \$75.00 flat. Minimum shipment: 1 pallet.

Destination	First Pallet	Destination	First Pallet
ACC	\$85	Elora	\$475
N. Mississauga	\$100	Kitchener	\$475
Toronto	\$130	Orillia	\$495
S. Mississauga	\$160	Stratford	\$575
Vaughan	\$165	Collingwood	\$620
Oakville	\$170	Thornbury	\$650
Caledon	\$260	Peterborough	\$655
Burlington	\$260	London	\$745
Markham	\$270	Belleville	\$785
Richmond Hill	\$270	Owen Sound	\$815
Stoney Creek	\$285	Huntsville	\$875
Ancaster	\$295	Picton	\$915
Ajax	\$300	Sarnia	\$1,045
Hamilton	\$300	Kingston	\$1,060
Grimsby	\$310	Harrow	\$1,235
Cambridge	\$370	North Bay	\$1,270
Oshawa	\$370	Windsor	\$1,325
Guelph	\$370	Sudbury	\$1,480
Brantford	\$415	Ottawa	\$1,675
Paris	\$430	Montreal	\$2,125
St. Catharines	\$430	Sault Ste. Marie	\$2,535
Barrie	\$435	Quebec City	\$3,030
Niagara	\$475		

Appendix B: Design Decisions

The following decisions were made during the specification of this framework and are recorded here for reference.

Decision	Rationale
Stem allocation method	Time-proportional. Stops consuming more route time bear more overhead. Reflects actual resource usage.
Fixed truck allocation method	Pallet-proportional. Balances cost against revenue since pallets drive revenue. Prevents low-pallet stops from being structurally over-charged.
Dwell cost rate	Operator rate only (\$0.7462/min). Asset rate excluded because the truck is stationary. Enables clean Dock SNI isolation.
Dwell prediction baseline	10 minutes base + 5 minutes per pallet. Universal starting point; to be calibrated against actual data after 90 days of operation.
Pricing schedule direction	Bidirectional. Pickups and deliveries priced identically. The truck makes the same round-trip regardless of freight direction.
Pricing cost basis	Round-trip transit cost plus pallet-proportional fixed truck share, divided by 0.60. Dwell excluded. Engineered for 40% margin on standalone delivery.
Revenue model	Each customer invoiced as standalone round-trip delivery regardless of route density. Multi-stop economics accrue entirely to ShipNoble.
Minimum shipment	1 pallet. Any freight occupying a pallet position is billed as a full pallet.
Multi-consignment split	Pallet-weighted. When customers share a stop, each bears cost proportional to their pallet count.
Pallet cost split	Revenue-proportional within a stop. First pallet (higher price) bears proportionally more cost than additional pallets (\$75).
Internal transfer costs	Allocated to the requesting department. Not absorbed into Outbound Manifest SNIs.
Linehaul treatment	Impact Multiplier adjusts cost by backhaul utilization. Net cost allocated to destination department.
Error cost	Not applicable. Single-CE damage events are operationally negligible.
Sequence deviations	System auto-repopulates manifest with actual order and recalculates Scheduled SNI. Variance always compares matching sequences.
Geographic hierarchy	45 Destination SNIs based on pricing schedule tiers. No rolled-up sector layer; pricing normalizes for distance.
Operator SNI normalization	Normalized against scheduled margin magnitude for cross-route comparability.
Shipper SNI window	Rolling 30-day, pooled calculation (sum revenue, sum costs, compute once).
DMS as authority	DMS system is the authoritative source for all Realized state data, not ELD hardware.

Appendix C: Constants Reference

Constant	Value	Derivation
Asset Rate	\$0.55 / km	Per-km variable cost of operating the Isuzu
Operator Rate	\$0.7462 / min	\$44.77/hr ÷ 60 minutes
Fixed Truck Cost (daily)	\$156.59	\$3,445/month ÷ 22 operating days (Mon–Fri)
Fixed Truck Cost (monthly)	\$3,445.00	Insurance + depreciation + licensing + parking + maintenance
Additional Pallet Rate	\$75.00 flat	Per additional pallet at same stop, all destinations
Feeder Sweep Piece Rate	\$2.00 / CE	Revenue per Case Equivalent collected on inbound sweeps
Dwell Base Time	10 minutes	Arrival, check-in, paperwork, departure (pre-calibration)
Dwell Per-Pallet Time	5 minutes	Physical unloading per pallet (pre-calibration)
Operating Days per Month	22	Standard Monday–Friday
Target Margin / SNI	40% / 1.0	Pricing divisor: 0.60 (complement of 40%)
Pricing Assumed Stops	2	Baseline for F_truck share in pricing; varies by tier